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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

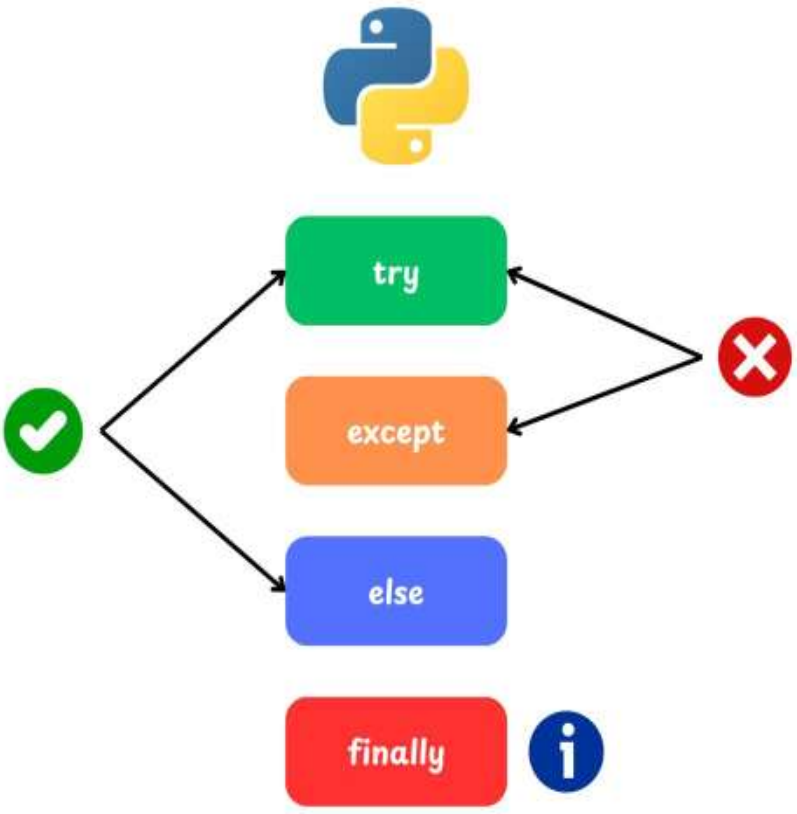
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 37: Exception Handling in Python



Learning Objectives

By the end of this lesson, students will be able to:

- Understand what an **error** is.
- Understand what an **exception** is.
- Identify common Python exceptions.
- Understand try and except.
- Use else and finally.
- Handle simple errors in programs.
- Build a small **Safe Calculator** project.

1. What is an Error?

An **error** is a problem in a computer program.

For example, imagine asking a calculator:

"What is 10 divided by 0?"

Mathematically, this is not allowed.

Python will report an error.

```
print(10 / 0)
```

Output:

```
ZeroDivisionError
```

2. What is an Exception?

An **exception** is a problem that happens while a Python program is running.

Instead of allowing the program to stop suddenly, we can **handle the problem**.

Think about this:

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🚗 A car is moving

⬇️

🚧 Obstacle appears

⬇️

🛑 Driver applies brakes

⬇️

🚗 Car safely stops

Exception handling works similarly:

Python Program

↓

Problem?

↓

try block

↓

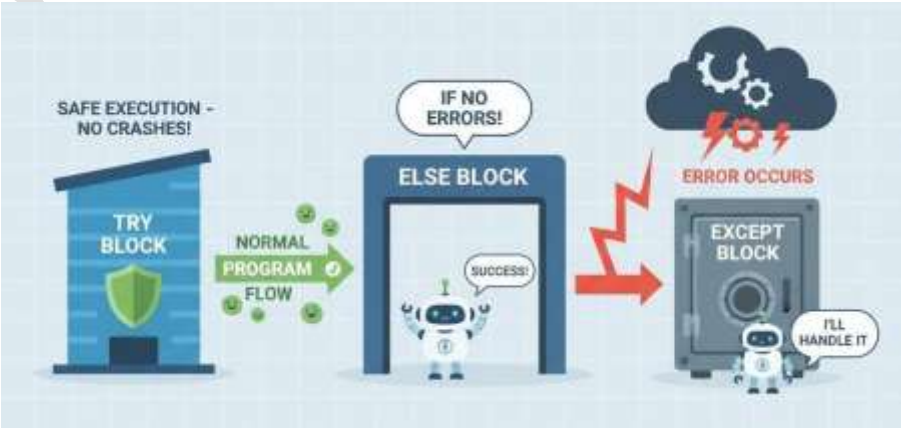
Error occurs

↓

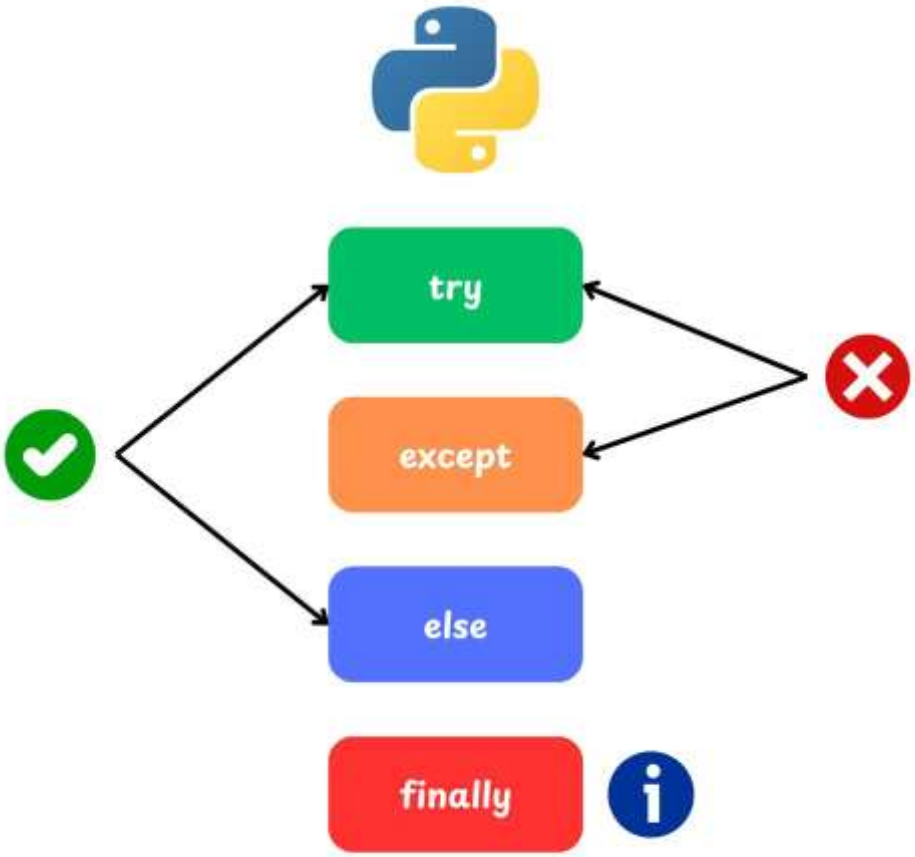
except handles it

↓

Program continues



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TRY The code that might cause an exception

```
try:
  # Code
```

EXCEPT The code to run if an exception occurs

```
except [ExceptionType]:
  # Code
```

FINALLY The code that will run no matter what

```
finally:
  # Code
```

3. Why Do We Need Exception Handling?

Without exception handling:

```
number = int(input("Enter a number: "))
print(10 / number)
```

If the student enters:

0

Python may stop the program with:

ZeroDivisionError

With exception handling, we can show a friendly message:

⚠️ You cannot divide by zero.

Please try again.

This makes programs:

- Safer
- Friendlier
- Easier to use
- Less likely to stop suddenly

4. The try Statement

The try block contains code that **might cause an error**.

Example:

```
try:
    number = int(input("Enter a number: "))
```

We tell Python:

"Try to execute this code."

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5. The except Statement

except tells Python what to do if an error occurs.

Basic Structure

```
try:
    # Code that may cause an error
except:
    # What to do if an error happens
```

Example:

```
try:
    number = int(input("Enter a number: "))
    print(number)
except:
    print("Please enter a number.")
```

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Practical Session

Practical 1 – Safe Number Input

```
try:
    number = int(input("Enter a number: "))
    print("You entered:", number)

except:
    print("Please enter a valid number.")
```

If the user enters:

25

Output:

You entered: 25

If the user enters:

hello

Output:

Please enter a valid number.

Common Python Exceptions

Python has different types of exceptions.

Exception	Meaning
ValueError	Wrong type of value
ZeroDivisionError	Dividing by zero
IndexError	Wrong list index
KeyError	Missing dictionary key
TypeError	Wrong type of operation
FileNotFoundError	File does not exist
NameError	Variable does not exist

For Grade 7–8, focus mainly on:

- ValueError
  ZeroDivisionError
  IndexError
  FileNotFoundError

7. ValueError

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A ValueError occurs when we give Python an inappropriate value.

Example:

age = int("hello")

Python cannot convert "hello" into an integer.

So:

ValueError

Handling ValueError

```
try:
    age = int(input("Enter your age: "))
    print("Your age is:", age)

except ValueError:
    print("Please enter numbers only.")
```

Input:

Enter your age: abc

Output:

Please enter numbers only.

8. ZeroDivisionError

Example:

print(10 / 0)

This causes:

ZeroDivisionError

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Handling ZeroDivisionError

```
try:
    number = int(input("Enter a number: "))
    result = 10 / number
    print(result)

except ZeroDivisionError:
    print("You cannot divide by zero.")
```

9. IndexError

Remember Python Lists?

```
fruits = ["Apple", "Mango", "Banana"]
```

Indexes are:

Apple → 0

Mango → 1

Banana → 2

If we write:

```
print(fruits[5])
```

there is no item at index 5.

Python gives:

IndexError

Handling IndexError

```
fruits = ["Apple", "Mango", "Banana"]
```

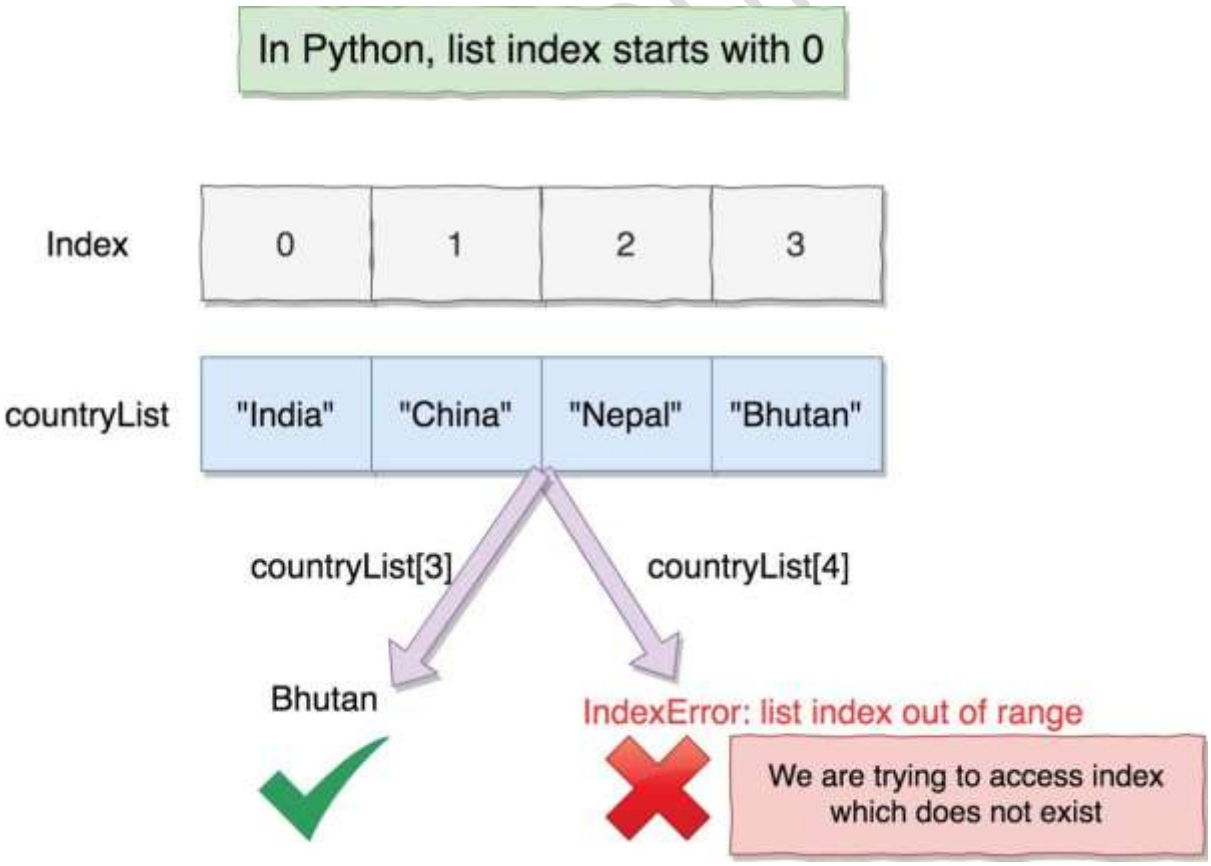
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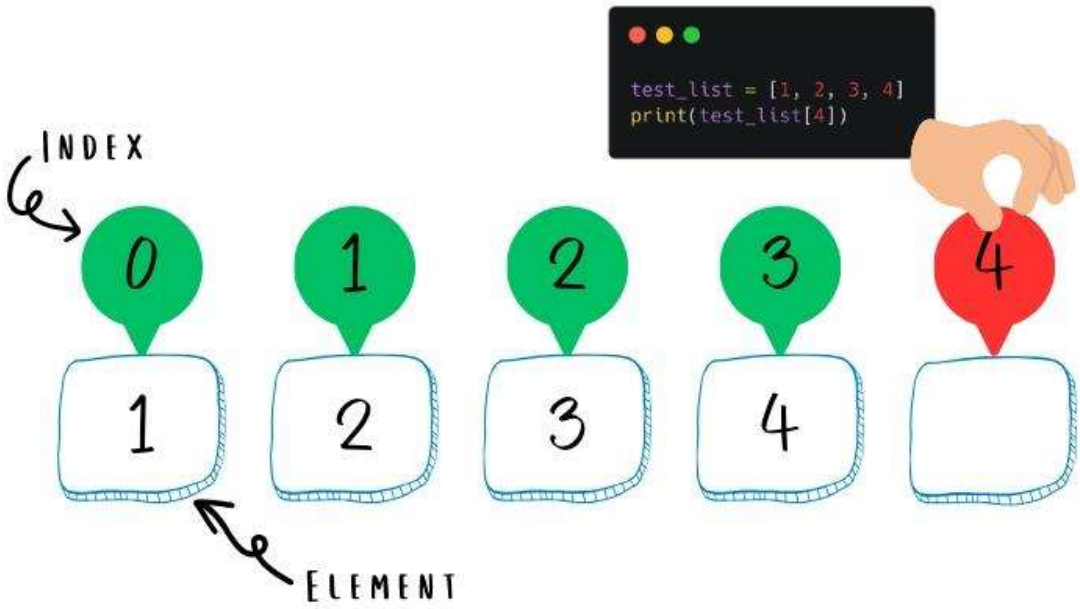
```
try:
    print(fruits[5])

except IndexError:
    print("That item does not exist.")
```

Output:

That item does not exist.






```
numbers = [10, 20, 30, 40]
index_variable = 5

print(numbers[index_variable])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback
Cell In[4], line 4
      1 numbers = [10, 20, 30, 40]
      2 index_variable = 5
----> 4 print(numbers[index_variable])

IndexError: list index out of range
```

10. FileNotFoundError

Suppose we try to open a file that doesn't exist:

file = open("school.txt", "r")

If school.txt isn't available, Python may produce:

FileNotFoundError

We can handle it:

```
try:
    file = open("school.txt", "r")
    print(file.read())
    file.close()

except FileNotFoundError:
    print("The file was not found.")
```

11. Handling Multiple Exceptions

Sometimes a program can have different problems.

Example:

```
try:
    number = int(input("Enter a number: "))
    result = 100 / number
    print(result)

except ValueError:
    print("Please enter a number.")

except ZeroDivisionError:
    print("Zero is not allowed.")
```

Now Python can handle both situations.

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12. The else Block

else runs when **no exception occurs**.

Structure:

```
try:
    # risky code

except:
    # error

else:
    # no error
```

Example:

```
try:
    number = int(input("Enter a number: "))

except ValueError:
    print("Invalid number.")

else:
    print("Great! You entered:", number)
```

If input is:

25

Output:

Great! You entered: 25

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13. The finally Block

finally runs **whether an error happens or not**.

Structure:

```
try:
    # code
except:
    # error handling
finally:
    # always runs
```

Example:

```
try:
    number = int(input("Enter a number: "))
    print(number)

except ValueError:
    print("Invalid input.")

finally:
    print("Program finished.")
```

Easy Way to Remember

TRY

↓

"Try this code."

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EXCEPT

↓

"Something went wrong."

ELSE

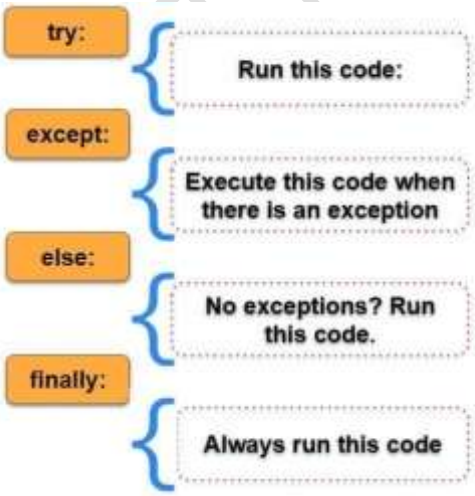
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"Everything worked!"

FINALLY

↓

"Do this at the end."



Practical 2 – Safe Calculator

Let's create a calculator that doesn't crash when the user enters invalid data.

```
try:
    a = int(input("Enter first number: "))
```

```
    b = int(input("Enter second number: "))

    result = a / b

    print("Result:", result)

except ValueError:
    print("Please enter numbers only.")

except ZeroDivisionError:
    print("You cannot divide by zero.")

else:
    print("Calculation completed successfully.")

finally:
    print("Thank you for using the calculator!")
```

Example 1

Input:

Enter first number: 20

Enter second number: 5

Output:

Result: 4.0

Calculation completed successfully.

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Thank you for using the calculator!

Example 2

Input:

Enter first number: 20

Enter second number: 0

Output:

You cannot divide by zero.

Thank you for using the calculator!

Example 3

Input:

Enter first number: twenty

Output:

Please enter numbers only.

Thank you for using the calculator!

Practical 3 – Guessing Game

```
secret = 7

try:
   guess = int(input("Guess a number: "))

   if guess == secret:
      print(" 🎉 Correct!")
```

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```
    else:
        print("❌ Try again!")

except ValueError:
    print("Please enter a number.")
```

Practical 4 – Marks Program

```
try:
    marks = int(input("Enter your marks: "))

    if marks >= 90:
        print("Grade A+")
    elif marks >= 75:
        print("Grade A")
    elif marks >= 60:
        print("Grade B")
    elif marks >= 40:
        print("Grade C")
    else:
        print("Need more practice.")

except ValueError:
    print("Please enter marks as a number.")
```

Practical 5 – List Safe Access

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```
subjects = ["Math", "Science", "English"]


try:
    position = int(input("Enter index number: "))
    print(subjects[position])


except ValueError:
    print("Please enter a number.")


except IndexError:
    print("That index does not exist.")
```

14. What Happens When an Exception Occurs?

Consider:

```
try:
    number = int(input("Enter number: "))
    print(10 / number)


except ZeroDivisionError:
    print("Cannot divide by zero.")
```

Flow:

User enters number

↓

Python executes TRY

↓

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Is there an error?

↙ ↘

YES NO

↓ ↓

EXCEPT Continue

↓ ↓

Message Program

15. Why Should We Handle Exceptions?

Imagine a game.

Player enters something incorrectly.

Without exception handling:

💥 PROGRAM CRASHED!

With exception handling:

⚠️ Invalid input.

Please try again.

Therefore:

Exception handling helps programs respond to unexpected situations safely.

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16. Common Mistakes

Mistake 1 – Incorrect indentation

❌

```
try:
number = int(input("Number: "))
except:
```

```
print("Error")
```

✅

```
try:
    number = int(input("Number: "))
except:
    print("Error")
```

Python uses indentation to understand blocks.

Mistake 2 – Wrong exception name

❌

```
except Valueerror:
```

✅

```
except ValueError:
```

Python exception names are **case-sensitive**.

17. Good Practice

For beginners, avoid using a completely empty except when you know the specific problem.

Instead of:

```
try:
    number = int(input("Number: "))
except:
    print("Error")
```

Prefer:

```
try:
```

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```
    number = int(input("Number: "))
except ValueError:
    print("Please enter a valid number.")
```

This tells us exactly what problem we are handling.

Classroom Activity

🚦 **"Error Detective"**

Give students these programs and ask:

Program A

```
print(10 / 0)
```

Question: What exception can occur?

✅ ZeroDivisionError

Program B

```
number = int("hello")
```

Question: What exception can occur?

✅ ValueError

Program C

```
animals = ["Cat", "Dog"]
print(animals[5])
```

Question: What exception can occur?

✅ IndexError

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Activity – Fix the Program

Give students this code:

```
try:
    age = int(input("Enter age: "))
    print("Age:", age)

except ValueError:
    print("Wrong input!")
```

Ask students to test:

12

Then:

abc

Students should observe the different results.

Mini Project: Safe Calculator

Project Goal

Create a calculator that can safely handle:

- Addition
- Subtraction
- Multiplication
- Division
- Invalid numbers
- Division by zero

Code

```
try:
```

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```
    first = float(input("Enter first number: "))
    second = float(input("Enter second number: "))

    operation = input("Choose +, -, *, /: ")

    if operation == "+":
        print("Answer:", first + second)

    elif operation == "-":
        print("Answer:", first - second)

    elif operation == "*":
        print("Answer:", first * second)

    elif operation == "/":
        print("Answer:", first / second)

    else:
        print("Unknown operation.")
except ValueError:
    print(" ⚠️ Please enter numbers only.")

except ZeroDivisionError:
    print(" ⚠️ Cannot divide by zero.")
```

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```
finally:
    print("Calculator closed.")
```

Challenge Project

Safe Quiz Program

Create a quiz with:

- 5 questions
- User answers
- Score calculation
- Invalid input handling

Example:

```
try:
    answer = int(input("What is 2 + 2? "))

    if answer == 4:
        print("Correct! 🎉 ")
    else:
        print("Wrong answer.")

except ValueError:
    print("Please enter a number.")

finally:
    print("Quiz question completed.")
```

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Exception Cheat Sheet

Exception	Simple Meaning	Example
ValueError	Invalid value	int("hello")
ZeroDivisionError	Division by zero	10 / 0
IndexError	Invalid list position	list[10]
KeyError	Missing dictionary key	student["age"]
TypeError	Wrong data type	"5" + 2
FileNotFoundError	File missing	open("abc.txt")
NameError	Unknown variable	print(score)

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Quiz

1. What is an exception?

A) A type of computer
B) A problem that occurs while a program runs
C) A keyboard button
D) A folder

✅ **Answer: B**

2. Which keyword is used to test code that may cause an exception?

A) check

B) try

C) test

D) error

✅ **Answer: B**

3. Which keyword handles an exception?

A) except

B) handle

C) error

D) catch

✅ **Answer: A**

4. Which exception occurs when dividing by zero?

A) ValueError

B) IndexError

C) ZeroDivisionError

D) NameError

✅ **Answer: C**

5. Which exception can occur when converting "hello" to an integer?

A) ValueError

B) IndexError

C) KeyError

D) FileError

✅ **Answer: A**

6. Which block runs when there is no exception?

A) except

B) else

C) error

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D) catch

✅ **Answer: B**

7. Which block normally runs at the end whether an exception occurs or not?

A) try

B) else

C) finally

D) stop

✅ **Answer: C**

Homework

Task 1

Write a program that asks for age and safely handles invalid input.

Task 2

Write a program that divides two numbers and handles division by zero.

Task 3

Create a list of 5 fruits and safely ask the user for an index.

Task 4 ⭐

Create a **Safe Marks Calculator** that:

1. Takes marks as input.
2. Calculates the grade.
3. Handles invalid input.
4. Displays a friendly error message.

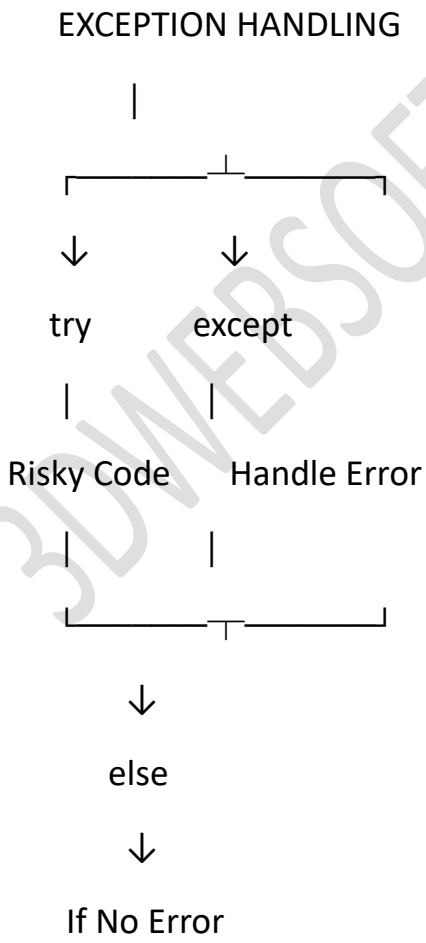
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Learning Check

Students should now be able to explain:

What is an error?
What is an exception?
Why do we use exception handling?
What does try do?
What does except do?
What does else do?
What does finally do?
What is ValueError?
What is ZeroDivisionError?
What is IndexError?

One-Minute Revision



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↓

finally

↓

Always Runs

Final Practical Task

Ask students to build a **"Student Safe Calculator"** that accepts two marks and calculates their average.

The program must:

- Take two marks.
- Calculate the average.
- Handle non-number input.
- Display a friendly error message.
- Use try, except, and finally.

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